

Bill of Material

The parts come from the McMaster-Carr database [1], Vevor [2] and Cimperial® [3]. The custom parts are made out of Aluminum.

#	Part Number	Description	Vendor	Quantity
1	6627T92	NEMA 17 Stepper Motor, 0.1067 hp	McMaster-carr	4
2	59585K14	Two-piece support-rail shaft: 20mm diameter, 1200mm long	McMaster-carr	2
3	95072A371	Cylindrical Precision Acme Externally Threaded Nut, 1.125"	McMaster-carr	5
4	99030A077	Carbon Steel Precision Acme Lead Screw, 800mm long, 0.75" diameter	McMaster-carr	2
5	2780T45	High Load Ball Bearing, 0.75" shaft diameter	McMaster-carr	6
6	9338T113	Mounted Linear Ball bearing for support rail shaft, 20mm shaft diameter, 54mm long	McMaster-carr	4
7	63255K49	Fixed Alignment High-Temp Linear ball bearing, 20mm shaft diameter	McMaster-carr	6
8	4147N187	Set Screw Precision Flexible Shaft Coupling, spiral-cut, 25.4mm long. 5mm shaft diameter	McMaster-carr	4
9	2349K792	Lubricated Ball Bearing, 17mm shaft diameter	McMaster-carr	4
10	4KWER20	4kW Air Cooled Spindle motor, CNC Grind Engraving	Vevor	1
11	4390N178	Flanged Thin Ball Bearing, 25mm shaft diameter, 7mm Wide	McMaster-carr	1
12	6627T105	NEMA 17 Position-Control DC Motor	McMaster-carr	1
13	1060CF	Metal work flood Coolant	Cimperial®	1
Custom Aluminum Parts				

14	243DP01	Device Casing		1
15	243DP02	Device Casing 2		1
16	243DP03	Support Bars for x-axis, 635mm long		2
17	243DP04	Support Bars for z-axis, 260mm long		2
18	243DP05	Bracket Mounting in x-axis motor, 45.5mm long		2
19	243DP06	Bracket Mounting in y-axis motor, 32.5mm long		2
20	243DP07	400mm x 400mm Board		1
21	243DP08	Rotary Axis Clamps, 75mm long		4
22	243DP09	Circular Tool holder with 18 slots		1
23	243DP10	4th Axis Shaft 100mm long, 17mm diameter		1
24	243DP11	4th Axis Shaft 70mm long, 17mm diameter		1
25	243DP12	4th axis Pulley, 8mm diameter		1
26	243DP13	4th axis Pulley, 20mm diameter		1
27	243DP14	Cone Shaped 4th Axis Fixed End Holder		1
28	243DP15	Rotating Cylinder 4th Axis End Holder, 120mm diameter		1
29	243DP16	Stair-shaped Clamps on 4th Axis Rotary End		4
30	243DP17	Fixed End Casing, 60mm x 100mm		1
31	243DP18	Rotary End Casing, 70mm x 100mm		1
32	243DP19	Casing Top Plate, 60mm x 100mm		1
33	243DP20	Casing Top Plate, 70mm x 100mm		1
34	243DP21	Spindle Support Body Mounted on x-axis		1

35	243DP22	Spindle Support Body Mounted to Spindle		1
36	243DP23	Tool Collet		1
37	243DP24	Pan Cross Head, M3.5 x 0.6 x 10 Type		8
38	243DP25	Pan Cross Head, M2 x 0.4 x 13 Type		4
39	243DP26	Pan Cross Head, M2 x 0.4 x 10 Type		8
40	243DP27	Pan Cross Head, M2 x 0.4 x 20 Type		4
41	243DP28	Pan Cross Head, M6 x 1.0 x 10 Type		32
42	243DP29	Pan Cross Head, M3 x 0.5 x 16 Type		10
43	243DP30	Pan Cross Head, M3 x 0.5 x 25 Type		9

[1] McMaster-carr, “Fastening and Joining,” in mcmaster.com, [Online]. Available: <https://www.mcmaster.com/>[Accessed: Dec - 2021]

[2]Vevor, “Vevor CNC 4kw air-cooled spindle motor ER20 Air cooled mill grind engraving,” vevor.ca, [Online]. Available: https://vevor.ca/products/vevor-cnc-4kw-air-cooled-spindle-motor-er20-air-cooled-mill-grind-engraving?gclid=CjwKCAiA1uKMBhAGEiwAxzvX95eZ5WjxujGLrkWGr51E9rnElp7t9jI7-oOzE944ixDNiF4tz4IcChoC8AMQAvD_BwE [Accessed: Dec - 2021]

[3] Cimcool, “Cimperial®” in cimcool.net [Online] Available: <http://www.cimcool.net/cimperial.html> . [Accessed: Dec - 2021]